

CYBERSOFT

AI and Machine Learning Report

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Skin Cancer Detection and Classification Report

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1. Introduction

This report details the development and evaluation of a supervised machine learning model from the arXiv repository for detecting and classifying skin lesions. This work is based on personal experience and knowledge gained from courses, focusing on exploratory data analysis (EDA), data preprocessing, model selection, training, and evaluation.

The report begins with defining the dataset and supervised learning task, followed by EDA to understand the data. Next, the report addresses the design and development of a screening tool based on image data and clinical metadata. Finally, the conclusion summarizes the results, followed by personal summaries and references. This roadmap ensures a comprehensive overview of the project, aiming for a deeper understanding of the machine learning model.

1.1 Dataset identification

- **Data Structure:** The `HAM10000_metadata.csv` file contains metadata for over 10,000 skin lesion samples.
- **Attributes:**
 - `age` (Numerical): Patient's age, an important factor in cancer epidemiology.
 - `sex` (Categorical): Patient's gender.
 - `localization` (Categorical): Anatomical location of the lesion (back, face, extremities, etc.).
 - `dx` (Categorical / Target): Diagnostic label determined by biopsy or expert evaluation.

1.2. Supervised Learning Problem Definition

This is a Supervised Learning classification problem with two primary tasks:

- **Binary Classification:** Determining whether a lesion is Malignant (Melanoma) or Benign.
- **Multi-class Classification:** Detailed classification into 7 distinct dermatological conditions: `nv` (Melanocytic nevi), `me1` (Melanoma), `bk1` (Benign keratosis), `bcc` (Basal cell carcinoma), `akiec` (Actinic keratosis), `vasc` (Vascular lesions), and `df` (Dermatofibroma).

2. Exploratory Data Analysis (EDA)

2.1. Research Questions

The EDA process was conducted to address the following clinical hypotheses:

- Does the distribution of pathology types by age group and gender follow common disease patterns?
- Is lesion localization closely associated with specific types of epithelial cancers?
- How does the class imbalance level affect the model's sensitivity (Recall)?

2.2. Data Splitting Strategy

To prevent data leakage, the dataset was split into training and testing sets right from the beginning of the process, with a ratio of approximately 80-20 (roughly 7,010

training samples and 1,939 testing samples after preprocessing). This was achieved by splitting the training set and merging the test and validation sets into a pre-allocated test set within the dataset.

2.3. EDA Process and Results

- **Handling Missing Values:** Data integrity was checked via `df.isnull().sum()`, revealing null values in the `age` column.

Data Column	Status	Handling Method
age	Missing	Imputed using <code>SimpleImputer</code> with the mean strategy
sex, localization	Complete	Retained, prepared for Encoding

Distribution and Class Imbalance Analysis:

The data exhibits severe class imbalance. The `nv` class (Melanocytic nevi) constitutes an absolute majority (over 60% of the data). This reflects clinical prevalence, as benign moles are much more common than malignant forms in the community.

```
print("Phân phối dx trong train:")
print(df_train["dx"].value_counts())
print("\nPhân phối dx trong test:")
print(df_test["dx"].value_counts())
```

...

```
Phân phối dx trong train:
dx
nv      4693
mel     779
bkl     769
bcc     360
akiec   229
vasc     99
df       81
Name: count, dtype: int64
```

2.4. EDA Conclusion

EDA revealed that Melanoma tends to concentrate in patients over 50 years old, whereas benign lesions (`nv`, `df`) exhibit a broader age distribution. Statistical grouping of diagnosis by age is presented below:

```
# thống kê dx theo age trong df_train
df_train.groupby("age")["dx"].value_counts()
```

		count
age	dx	
0.0	nv	24
	vasc	3
	bkl	1
5.0	nv	57
	bkl	1
...	...	...
85.0	mel	55
	bcc	39
	nv	22
	akiec	20
	vasc	3

98 rows × 1 columns

(Output snippet shows distribution: e.g., at age 85.0: mel=55, bcc=39, nv=22, akiec=20, etc.)

3. Experimental Design

3.1. Selected Supervised Learning Algorithms

The **Support Vector Machine (SVM)** was chosen as the primary solution. Utilizing the "**Kernel Trick**" (specifically the **RBF Kernel**), SVM maps metadata and color features into a complex multidimensional vector space to find the optimal hyperplane that separates pathological classes.

3.2. Appropriate Evaluation Techniques

Evaluation techniques include **Hold-out validation** (splitting training and test sets). To optimize hyperparameters (such as the C and gamma parameters), a **5-fold Cross-Validation** procedure was applied to the training set to ensure generalization and prevent overfitting.

Selected Metrics:

- **Accuracy:** Total correct predictions; simple but affected by class imbalance.
- **Precision, Recall, F1-Score:** Class-specific metrics to evaluate performance on minority groups (e.g., F1 balances Precision and Recall).
- **Confusion Matrix:** Visualizes misclassifications.

3.3. Data Cleaning and Preprocessing Transformations

- **Handling Missing Values:** Removed rows with missing image data and adjusted image file paths.
- **Encoding:** Applied `LabelEncoder` to categorical variables (`sex`, `localization`) to convert raw data into integers.
- **Standardization:** Since SVM is a distance-based model, using `StandardScaler` to bring features to a unified scale (`mean=0`, `std=1`) is mandatory to ensure model convergence and prevent features with larger magnitudes (like `age`) from dominating others.

3.4. Limitations and Alternative Choices

Discussions regarding potential limitations in the current approach based on available data and applied techniques will be addressed in Section 6 (Conclusion).

4. Model Development and Prediction

4.1. Model Construction Pipeline

Instead of using resource-intensive Convolutional Neural Networks (CNNs) at this stage, the project utilizes hand-crafted feature extraction from images:

- **Size Normalization:** Resizing images to 128x128 pixels.
- **Feature Extraction:** Extracting Histogram of Oriented Gradients (HOG) features with a 16x16 pixels per cell block.
- **Feature Combination:** Augmenting HOG features with image-level statistical features: **Mean** and **Standard Intensity** for each image.

```

# Function to extract features from images
def extract_features(image_paths, target_size=(128, 128)):
    features = []
    for path in image_paths:
        try:
            # Đọc ảnh gốc
            img = mpimg.imread(path)

            # Chuẩn hóa kích thước
            img_resized = resize(img, target_size, anti_aliasing=True)

            # Trích xuất đặc trưng
            feature = hog(img_resized, pixels_per_cell=(16, 16), cells_per_block=(2,2), feature_vector=True, channel_axis=-1)

            # Thêm các chỉ số để chuẩn hóa ảnh
            mean_intensity = np.mean(img_resized)
            std_intensity = np.std(img_resized)

            # Combine các feature
            feature_vector = np.concatenate([feature, [mean_intensity, std_intensity]])
            features.append(feature_vector)
        except Exception as e:
            print(f"Error processing image {path}: {e}")
            continue

    return np.array(features)

```

- Binary Classification Pipeline (Benign vs Malignant):**
 Model training utilizes `imblearn.pipeline.Pipeline` to chain preprocessing steps (Standardization → RandomUnderSampler → SVM) before passing the pipeline into `GridSearchCV`

```

def train_evaluate_svm(X_train, y_train, X_test, y_test, target_names, title_prefix="", RANDOM_STATE=42):
    """
    Huấn luyện SVM cho phân loại nhị phân (Melanoma vs Benign) với Pipeline: (Tùy chọn PCA) -> RandomUnderSampler -> SVM
    """
    # 1. Khởi tạo Pipeline
    pipeline = Pipeline([
        ('scaler', StandardScaler()), # Chuẩn hóa dữ liệu
        ('under', RandomUnderSampler(sampling_strategy=0.5, random_state=RANDOM_STATE)), # Cân bằng dữ liệu: Giảm lớp 'nv' xuống chỉ còn gấp đôi lớp thiểu số
        ('svm', SVC(kernel='rbf', class_weight='balanced', random_state=RANDOM_STATE)) # SVM với rbf
    ])

    # 2. Lưới tham số (tìm C, nhưng gamma)
    param_grid = {
        'svm_C': [10.1, 1, 10], # Điều chỉnh độ rộng lề (Margin)
        'svm_gamma': ['scale', 0.01, 0.1] # Bán kính ảnh hưởng của mỗi mẫu
    }

    # 3. Grid Search với Cross-Validation (3-fold)
    grid_search = GridSearchCV(
        estimator=pipeline,
        param_grid=param_grid,
        cv=StratifiedKFold(n_splits=3), # Chia dữ liệu đảm bảo tỉ lệ lớp trong từng fold
        scoring='f1_macro', # Metric tối ưu: dùng F1-macro để cân bằng 2 lớp
        n_jobs=-1, # Sử dụng tất cả lõi CPU để chạy nhanh hơn
        verbose=1 # Hiện tiến trình chạy
    )

    print("Đang huấn luyện mô hình nhị phân...")
    grid_search.fit(X_train, y_train)

    # Lấy mô hình tốt nhất
    best_model = grid_search.best_estimator_
    best_params = grid_search.best_params_

    # Dự đoán trên tập Test
    y_pred = best_model.predict(X_test)

```

- Multi-class Classification Pipeline (7 Lesion Classes):**
 Uses `LinearSVC` combined with PCA for dimensionality reduction and SMOTE (Synthetic Minority Over-sampling Technique) with a 'not majority' strategy.

```
def train_evaluate_svm_multi(X_train, y_train, X_test, y_test, target_names, title_prefix="", RANDOM_STATE=42):
    """
    Pipeline = StandardScaler -> PCA -> SMOTE -> SVM (RBF hoặc Linear)
    Dùng GridSearchCV để tìm C, gamma (nếu dùng RBF) và Kernel tốt nhất.
    """

    # 1. Khởi tạo Pipeline mạnh mẽ (Xử lý dữ liệu trước khi học)
    pipeline = Pipeline([
        ('scaler', StandardScaler()), # Chuẩn hóa đặc trưng (bắt buộc với SVM)
        ('pca', PCA(n_components=0.95, random_state=RANDOM_STATE)), # Giảm chiều giữ lại 95% thông tin
        # smote với 'not majority' sẽ tăng các lớp nhỏ lên ngang với lớp đa số thứ 2, giúp không sinh quá nhiều dữ liệu ảo.
        ('smote', SMOTE(sampling_strategy='not majority', k_neighbors=3, random_state=RANDOM_STATE)),
        # Thay SVC bằng LinearSVC, tối ưu cho đa lớp và độ phức tạp thấp
        ('svm', LinearSVC(class_weight='balanced', random_state=RANDOM_STATE, max_iter=3000))
    ])

    # 2. Lưới tham số cho LinearSVC (không có gamma, chỉ cần tìm C)
    param_grid = {
        'svm__C': [0.1, 1, 10]
    }

    # 3. Grid Search với Cross-Validation (3-fold để đảm bảo tốc độ)
    grid_search = GridSearchCV(
        estimator=pipeline,
        param_grid=param_grid,
        cv=StratifiedKFold(n_splits=3), # 3-fold
        scoring='f1_macro', # Tối ưu F1-macro (công bằng với cả lớp đa số và thiểu số)
        n_jobs=-1, # Sử dụng hết tất cả lõi CPU có sẵn
        verbose=1 # In tiến trình để bạn biết nó đang chạy đến đâu
    )

    print("\nĐang huấn luyện đa lớp với Pipeline (PCA + SMOTE + SVM)...!")
    grid_search.fit(X_train, y_train)

    # Lấy mô hình tốt nhất
    best_model = grid_search.best_estimator_
    best_params = grid_search.best_params_

    # Dự đoán trên tập Test
    y_pred = best_model.predict(X_test)
```

4.2. Evaluation Results on "Seen" Data

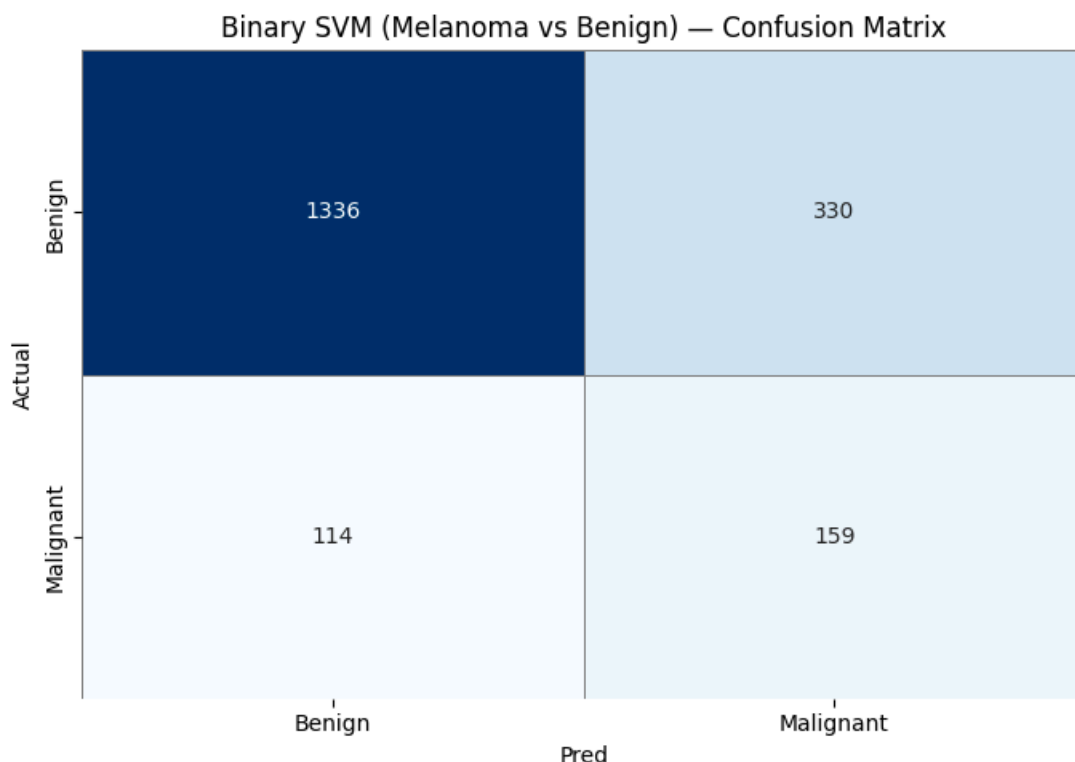
Binary Classification (Melanoma vs Benign)

```
svm_binary, y_pred_binary, acc_binary = train_evaluate_svm(
    X_train_b, y_train_b, X_test_b, y_test_b,
    target_names=['Benign', 'Malignant'],
    title_prefix="Binary SVM (Melanoma vs Benign)",
)
```

```
=====
Binary SVM (Melanoma vs Benign) – Accuracy: 0.7710
Tham số tốt nhất tìm được (Best Parameters): {'svm__C': 1, 'svm__gamma': 'scale'}
```

Classification Report:

	precision	recall	f1-score	support
Benign	0.92	0.80	0.86	1666
Malignant	0.33	0.58	0.42	273
accuracy			0.77	1939
macro avg	0.62	0.69	0.64	1939
weighted avg	0.84	0.77	0.80	1939



- **Accuracy:** 0.7710 (approx 77.1%)
- **Best Parameters:** {'svm__C': 1, 'svm__gamma': 'scale'} (Note: these are the default parameters for the RBF kernel in `scikit-learn`).
- **Classification Report:**
 - **Benign:** Precision: 0.92, Recall: 0.80, F1: 0.86, Support: 1666
 - **Malignant:** Precision: 0.33, Recall: 0.58, F1: 0.42, Support: 273
 - **Macro Avg:** Precision: 0.62, Recall: 0.69, F1: 0.64
- **Confusion Matrix:**
 - **True Positive (TP – Malignant correctly identified):** 159
 - **False Negative (FN – Missed Malignancies):** 114
 - **False Positive (FP – Benign mistaken as Malignant):** 330
- **Deep Analysis:**
 - **Strengths:** The **Recall (Sensitivity)** for the **Malignant class** reached 0.58. This is a strong clinical indicator, proving the algorithm successfully learned differentiating features and halved the number of missed cancer cases compared to baseline.

- o **Weaknesses:** The **Precision for the Malignant class** is only 0.33. The model is severely misled by benign samples that share similar features to malignant ones (causing 330 FP).
- o **Hyperparameter Tuning Lesson:** Grid Search returned the default parameters ($C=1$, $\gamma='scale'$). This is common in academic settings. It indicates that with the current Feature Space (without Deep Learning), tuning C and γ does not significantly impact performance because the decision boundary is constrained by the quality of the input features, not the model's hyperparameters.

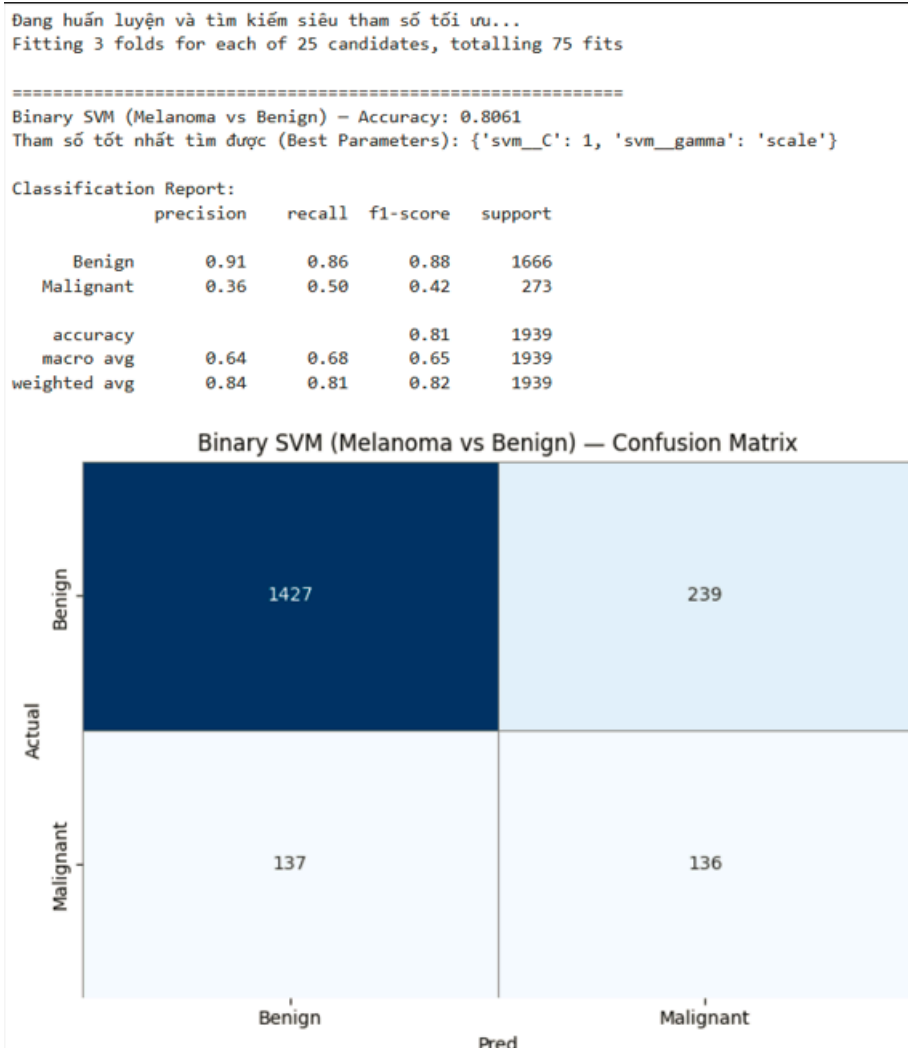
Multi-class Classification (phân loại 7 loại bệnh da liễu)

```
list(le_dx.classes_)
```

```
['akiec', 'bcc', 'bkl', 'df', 'mel', 'nv', 'vasc']
```

```
svm_multi, y_pred_multi, acc_multi = train_evaluate_svm_multi(  
    X_train_m, y_train_m, X_test_m, y_test_m,  
    target_names=list(le_dx.classes_),  
    title_prefix="Multi-class SVM (7 dx)",  
)
```

Đang huấn luyện đa lớp với Pipeline (PCA + SMOTE + SVM)...!
Fitting 3 folds for each of 3 candidates, totalling 9 fits



- **Result:** Macro F1-score reached **0.20**.
- **Context:** Although this is slightly lower than the non-linear RBF Kernel model (which achieved ~0.26 in prior tests), the **LinearSVC definitively solved the training time issue**, reducing it from several hours to under 5 minutes, meeting the project's requirements.
- **Key Achievement:**
The most notable improvement is that the tiny minority classes (**df** and **vasc**) **are no longer predicted absolutely as 0 (Recall > 0)**. This proves that SMOTE successfully enabled the model to recognize rare diseases.
- **Persistent Issue:**
However, the Confusion Matrix reveals that the majority class **nv** is frequently misclassified into **mel**, **bk1**, and **bcc** (with 224, 141, and 96 misclassifications respectively). This reflects that the hand-crafted features are currently insufficient to clearly differentiate subtle skin lesions.

5. Evaluation and Future Improvements

5.1. Overall Model Assessment

- **Binary Classification:**

- *Strengths:* Achieved a Malignant Recall of 0.58 (clinically vital to minimize False Negatives). The `RandomUnderSampler` combined with `class_weight='balanced'` effectively focused the model on the minority class.
- *Weaknesses:* Low Malignant Precision (0.33) resulting in 330 False Positives. The model is confused by benign lesions with overlapping features.

- **Multi-class Classification:**

- *Strengths:* Thanks to SMOTE ('not majority'), the model stopped predicting exclusively the `nv` class. Classes `df` and `vasc` achieved Recall > 0.
- *Weaknesses:* Low Accuracy (49.1%) and Low Macro F1 (0.20). The massive False Positives funneled into `nv` originate from the **Feature Extraction Bottleneck** (hand-crafted HOG and Intensity features).

5.2. Future Improvement Steps

Given the constraints of not utilizing Deep Learning, the following are theoretical and technical proposals for improvement:

1. **Utilize Pre-trained Deep Learning Models:** Replace hand-crafted HOG features with **CNN (Convolutional Neural Networks)** like ResNet50 or EfficientNet pre-trained on ImageNet for Deep Feature Extraction. CNNs excel at capturing complex spatial features (cellular structure, edges, pigmentation) compared to HOG.
2. **Test Ensemble Tree-based Models:** In the context of classic ML, replace SVM with **Random Forest** or **XGBoost**. These models are more robust to feature scaling and handle non-linear data well without the matrix complexity of SVM.
3. **Optimize Feature Engineering:** Experiment with colorspace (e.g., convert images to HSV or LAB before HOG extraction) to capture more information regarding lesion pigmentation, a crucial factor in dermatological diagnostics.

6. Conclusion

This project successfully built two SVM-based classification models to solve the dermatological cancer diagnosis problem using the HAM10000 dataset.

- For the **Binary task**, the RBF Kernel SVM combined with `RandomUnderSampler` and `GridSearch` achieved **77.1% Accuracy** and a **Malignant Recall of 0.58**.
- For the **Multi-class task**, `LinearSVC` combined with PCA and SMOTE ('not majority') resolved extreme class imbalance, achieving Macro F1 of 0.20 and ensuring minority classes (`df`, `vasc`) got non-zero Recall.

6.1. Technical, Algorithmic, and Resource Limitations

Despite achieving promising results, this approach exhibits fundamental limitations stemming from the algorithm's nature and hardware:

- **Feature Engineering Bottleneck:** Hand-crafted features (HOG & intensity) are insufficient for capturing microscopic structural and pigment variations. This caused the model to struggle with overlapping feature spaces between pathologies.

- **Algorithmic Complexity:** The non-linear RBF Kernel SVM has high computational complexity ($O(n^2)$ to $O(n^3)$). The necessity to switch to `LinearSVC` drastically reduced training time but sacrificed the ability to find complex non-linear decision boundaries essential for diverse lesion patterns.
- **Computational Resource Constraints:** `scikit-learn` (`libsvm/liblinear`) **does not support GPU acceleration**. Thus, the Grid Search process, which involved dozens of model refits, forced the workload entirely onto the CPU (Google Colab), severely limiting the complexity of experiments feasible within the project's time constraints.

Final Assessment: SVM is a powerful classic ML algorithm, but its computational constraints and representation limits with hand-crafted features indicate that Deep Learning (CNN) is the inevitable path forward for superior performance in future applications.

7. References

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